

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402073
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hsiao; Po-Wen

Control method of user equipment with adaptive system selection strategy

Abstract

The present invention provides a control method of an electronic device, wherein the control method includes the steps of: obtaining a first measurement result of frequencies of one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result.

Inventors:	Hsiao; Po-Wen (Hsinchu, TW)
Applicant:	MEDiatek INC. (Hsin-Chu, TW)
Family ID:	1000008777482
Assignee:	MEDiatek INC. (Hsinchu, TW)
Appl. No.:	17/844716
Filed:	June 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230026195 A1	Jan. 26, 2023

Related U.S. Application Data

us-provisional-application US 63221975 20210715

Publication Classification

Int. Cl.: H04W52/02 (20090101)

U.S. Cl.:

CPC H04W52/0225 (20130101);

Field of Classification Search

CPC: H04B (7/0695); H04B (7/0617); H04W (24/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0121194	12/2012	Heshmati	370/252	H04W 52/0258
2013/0316699	12/2012	Jheng	455/423	H04W 48/18
2015/0156713	12/2014	Song	N/A	N/A
2016/0013878	12/2015	Zhao	370/329	H04J 11/0023
2017/0070904	12/2016	Mali	N/A	N/A
2017/0094589	12/2016	Bhasin	N/A	N/A
2019/0239238	12/2018	Calabrese	N/A	N/A
2020/0329338	12/2019	Lovlekar	N/A	N/A
2020/0351066	12/2019	Cirik	N/A	H04L 5/0098
2021/0194563	12/2020	Zhang	N/A	H04W 24/10
2022/0015100	12/2021	Yerramalli	N/A	H04L 5/0023

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108668329	12/2017	CN	N/A
112243278	12/2020	CN	N/A

Primary Examiner: Ferris; Derrick W

Assistant Examiner: Choi; Won Jun

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/221,975, filed on Jul. 15, 2021. The content of the application is incorporated herein by reference.

BACKGROUND

(1) When an user equipment (UE) moves to a limited or no service area, such as a basement, the UE always triggers regularly recovery search for detecting a normal service according to 3rd Generation Partnership Project (3GPP) 36.304 and 38.304. The recovery search comprises the step of finding a suitable cell by trying all frequencies of all radio access technologies (RATS) that are supported by the UE, causing high power consumption. Therefore, if the UE is located in an area without normal service for a long time, the UE will continue to perform redundant recovery search regularly, resulting in high power consumption and the power of the UE will drop rapidly.

SUMMARY

- (2) It is therefore an objective of the present invention to provide a control method of the UE, which can use an adaptive system selection strategy to avoid the high power consumption caused by the conventional recovery search in the prior art.
- (3) According to one embodiment of the present invention, a control method of an electronic device is disclosed. The control method comprises the steps of: obtaining a first measurement result of frequencies of one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result.
- (4) According to one embodiment of the present invention, a circuitry of an electronic device is disclosed, wherein the circuitry is configured to perform the steps of: obtaining a first measurement result of frequencies of one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result.
- (5) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a communication system according to one embodiment of the present invention.
- (2) FIG. 2 shows a control method of UE based on measurement results of frequencies/channels of or more bands.
- (3) FIG. 3 is a diagram illustrating a control method of the UE according to a first embodiment of the present invention.
- (4) FIG. 4 shows steps of recovery search of the UE.
- (5) FIG. 5 is a diagram illustrating a control method of the UE according to a second embodiment of the present invention.
- (6) FIG. 6 is a diagram illustrating a control method of the UE according to a third embodiment of the present invention.

DETAILED DESCRIPTION

- (7) Certain terms are used throughout the following description and claims to refer to particular system components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .”. The terms “couple” and “couples” are intended to mean either an indirect or a direct electrical connection. Thus, if a first device couples to a second device, that connection may be through a direct electrical connection, or through an indirect electrical connection via other devices and connections.
- (8) FIG. 1 shows a communication system according to one embodiment of the present invention. As shown in FIG. 1, an UE **100** comprises a circuitry comprising an application processor for executing operations of an user application **110**, a middleware **120**, a transport layer **130**, a data link layer **140**, and circuitry further at least one wireless network module **150**. In this embodiment,

the UE **100** may be a portable electronic device such as a cell phone or a tablet, and the UE **100** can use the wireless network module **150** to establish a link with a cellular base station for a network system such as a fourth generation of broadband cellular network technology (hereinafter, 4G) system or a fifth generation of broadband cellular network technology (hereinafter, 5G) system, or a next generation of broadband cellular network technology and legacy network technology (hereinafter, 2G and 3G) system.

(9) As described in the background of the invention, when the UE **100** is taken into a limited or no service area, such as a basement, the UE **100** will trigger regularly recovery search for detecting a normal service according to the 3GPP specification. In order to lower the power consumption of the UE **100** when the UE **100** stays in the area without service for a long time, the embodiment shown in FIG. **2** provides a control method of the UE **100** to select an appropriate system selection procedure strategy according to a measurement result of frequencies/channels of one or more bands, wherein the system selection procedure strategies that can be used by the UE **100** may comprise different cell search strategies, different public land mobile network (PLMN) and RAT selection strategy, and/or different system capability control strategy.

(10) Referring to FIG. **2**, at a first time, the wireless network module **150** of the UE **100** obtains a first measurement result by performing a power scan operation for frequencies/channels of one or more bands, then at a second time, the wireless network module **150** of the UE **100** may obtain a second measurement result by performing the power scan operation for the frequencies/channels of one or more bands. Then, the first measurement result and the second measurement result are compared to determine a similarity between the second measurement result and the first measurement result, for determining the suitable system selection procedure strategy for the UE **100**. For example, if the second measurement result is similar to the first measurement result, the wireless network module **150** can determine that the UE **100** may be located in the same area, and the wireless network module **150** can use a first system selection procedure strategy at the next time. For example, if the UE **100** is located in the basement without service and the first measurement result and the second measurement are generated, in response to the second measurement result being similar to the first measurement result, the wireless network module **150** may not perform the cell search or the PLMN and RAT selection, or reduce the redundant cell search, to avoid performing actions that do not achieve the desired effect. In another example, if the UE **100** is without service at the first time and the first measurement result is generated, then the second measurement result is generated at the second time, in response to the second measurement result being not similar to the first measurement result, the wireless network module **150** may perform a normal recovery search because the UE **100** may move to an area having service now.

(11) FIG. **3** is a diagram illustrating a control method of the UE **100** according to a first embodiment of the present invention. As shown in FIG. **3**, in step **300**, the flow starts, and the UE **100** is connected to a cellular base station of a system. In Step **302**, the UE **100** cannot receive the signal from the cellular base station due to some reason, that is the UE **100** is out of wireless communication service, and the UE **100** starts to perform a recovery search and obtain a measurement result of frequencies/channels of one or more bands. Specifically, referring to FIG. **4**, initially, because the initial memory within the UE **100** stores the previous connection information (e.g., the information of the cellular base station of the 4G/5G system), the wireless network module **150** may determine if the frequency of the stored connection information is available for connecting to the cellular base station of the 5G system, if yes, the UE **100** can establish a link with the cellular base station, and the procedure of the recovery search is finished; and if not, the wireless network module **150** detect power of all the frequencies or part of the frequencies of several bands to obtain a measurement result of frequencies/channels, wherein these frequencies/channels comprise at least part of standalone architecture (SA) FR1 and FR2 bands of the 5G system. In this embodiment, the power of each frequency/channel in the measurement result can be received signal strength indicator (RSSI), a reference signal receiving power (RSRP), a

reference signal received quality (RSRQ), a signal to interference plus noise ratio (SINR), or any suitable power indicator. After obtaining the measurement result, the wireless network module **150** performs cell search to try to connect to the cellular base station of the 5G system based on the measurement result.

(12) Then, if the UE **100** cannot connect to the 5G system, the wireless network module **150** may determine if the frequency of the stored connection information is available for connecting to the cellular base station of the 4G system, if yes, the UE **100** can establish a link with the cellular base station of the 4G system, and the procedure of the recovery search is finished; and if not, the wireless network module **150** detect power of all the frequencies or part of the frequencies of several bands to obtain a measurement result of frequencies/channels, wherein these frequencies/channels comprise at least part of bands B1-B71 of the 4G system. In this embodiment, the power of each frequency/channel in the measurement result can be RSSI, RSRP, RSRQ, SINR, or any suitable power indicator. After obtaining the measurement result, the wireless network module **150** tries to connect to the cellular base station of the 4G system based on the measurement result. Similarly, the wireless network module **150** may also try to connect to the bands of the 3G or 2G system.

(13) It is noted that the flow of the recovery search shown in FIG. **4** is for illustrative purposes only, not a limitation of the present invention. In one embodiment that the UE **100** does not have the 5G service, the flow for generating the measurement result and performing cell search of the 5G system can be removed.

(14) In this embodiment, the measurement result of the 5G system and/or the measurement result of the 4G system can be temporarily stored in a buffer of the wireless network module **150**.

(15) In Step **304**, because the UE **100** cannot have service in the above recovery search, in a next recovery search, initially the wireless network module **150** determines if the frequency of the stored connection information is available for connecting to the cellular base station of the 5G system and/or the 4G system, if yes, the flow enters Step **308** to use the stored connection information to connect to the cellular base station, and the recovery search procedure is finished; and if not, the flow enters Step **310**.

(16) In Step **310**, the wireless network module **150** detects power of all the frequencies or part of the frequencies of several bands to obtain a measurement result of frequencies/channels, wherein these frequencies/channels comprise at least part of SA FR1 and FR2 bands of the 5G system and/or at least part of bands B1-B71 of the 4G system.

(17) In Step **312**, the wireless network module **150** determines the measurement result obtained in Step **310** is similar to the measurement result obtained in Step **302**, if yes, the flow enters Step **314**; and if not, the flow enters Step **316**. In this embodiment, the wireless network module **150** may use any suitable algorithm to determine the similarity between these two measurement results, such as calculating a distance between two measurement results, or using a suitable artificial intelligence (AI) model to calculate a difference between two measurement results. Then the wireless network module **150** can determine if the difference between two measurement results is within a range or lower than a threshold value to determine the similarity between these two measurement results. For example, if the difference between two measurement results is within the range or lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **310** is similar to the measurement result obtained in Step **302**; and if the difference between two measurement results is not within the range or not lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **310** is not similar to the measurement result obtained in Step **302**.

(18) In Step **314**, the wireless network module **150** determines that the UE **100** may be located in the same area without service, so the wireless network module **150** may not perform a full-band cell search for the 5G or 4G system. For example, the wireless network module **150** does not perform cell search on at least part of the SA FR1 and FR2 bands of the 5G system, and/or the

wireless network module **150** does not perform cell search on at least part of the bands B1-B71 of the 4G system.

(19) In one embodiment, the recovery search performed in Step **314** is band-independent, that is the wireless network module **150** does not perform cell search on the bands whose measurement result is with high similarity to the previous measurement result, but the wireless network module **150** may perform cell search on the bands whose measurement result is with low similarity to the previous measurement result.

(20) In addition, after the Step **314**, the flow enters Step **304** again, and the UE **100** performs the next recovery search and determines the similarity between the current measurement result and the previous measurement result(s) to determine a suitable system selection strategy.

(21) In Step **316**, the wireless network module **150** determines that the UE **100** may be moved to an area with service, so the wireless network module **150** may perform a cell search on all the possible frequencies of the 5G or 4G system.

(22) FIG. 5 is a diagram illustrating a control method of the UE **100** according to a second embodiment of the present invention. As shown in FIG. 5, in step **500**, the flow starts, and the UE **100** is connected to a cellular base station of a system such as the 5G system or the 4G system. In Step **502**, the UE **100** cannot receive the signal from the cellular base station due to some reason, that is the UE **100** is out of wireless communication service, and the UE **100** starts to perform a recovery search and obtain a measurement result of frequencies/channels of one or more bands. For example, the measurement result can be generated according to the above embodiment shown in FIG. 4.

(23) In this embodiment, the measurement result of frequencies/channels of one or more bands can be temporarily stored in a buffer of the wireless network module **150**.

(24) In Step **504**, because the UE **100** cannot have service in the above recovery search, in a next recovery search, initially the wireless network module **150** determines if the frequency of the stored connection information is available for connecting to the cellular base station of the 5G system and/or the 4G system, if yes, the flow enters Step **508** to use the stored connection information to connect to the cellular base station, and the recovery search procedure is finished; and if not, the flow enters Step **510**.

(25) In Step **510**, the wireless network module **150** detects power of all the frequencies or part of the frequencies of one or more bands to obtain a measurement result of frequencies/channels, wherein these frequencies/channels comprise at least part of SA FR1 and FR2 bands of the 5G system and/or at least part of bands B1-B71 of the 4G system.

(26) In Step **512**, the wireless network module **150** determines the measurement result obtained in Step **510** is similar to the measurement result obtained in Step **502**, if yes, the flow enters Step **514**; and if not, the flow enters Step **516**. In this embodiment, the wireless network module **150** may use any suitable algorithm to determine the similarity between these two measurement results, such as calculating a distance between two measurement results, or using a suitable artificial intelligence (AI) model to calculate a difference between two measurement results. Then, the wireless network module **150** can determine if the difference between two measurement results is within a range or lower than a threshold value to determine the similarity between these two measurement results. For example, if the difference between two measurement results is within the range or lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **510** is similar to the measurement result obtained in Step **502**; and if the difference between two measurement results is not within the range or not lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **510** is not similar to the measurement result obtained in Step **502**.

(27) In Step **514**, the wireless network module **150** determines that the UE **100** may be located in the same area without service, so the wireless network module **150** may not perform part of the PLMN and RAT selections. For example, if the previous connection information comprises a

specific PLMN and a specific RAT, the wireless network module **150** does not try to select the specific PLMN and the specific RAT.

(28) In Step **516**, the wireless network module **150** determines that the UE **100** may be moved to an area with service, so the wireless network module **150** performs the PLMN and RAT selections in the recovery search.

(29) FIG. **6** is a diagram illustrating a control method of the UE **100** according to a third embodiment of the present invention. As shown in FIG. **6**, in step **600**, the flow starts, and the UE **100** is connected to a cellular base station of a system such as the 5G system or the 4G system. In Step **602**, the UE **100** disables the 5G function due to some reason, that is the UE **100** does not try to connect to the cellular base station even if the 5G connection is available. For example, the UE **100** may be unable to connect to any cellular base station of the 5G system for a long time, so the wireless network module **150** actively disables the 5G function. In addition, during the recovery search performed by the UE **100**, the UE **100** obtains a measurement result of frequencies/channels of one or more bands. For example, the measurement result can be generated according to the above embodiment shown in FIG. **4**.

(30) In this embodiment, the measurement result of frequencies/channels of one or more bands can be temporarily stored in a buffer of the wireless network module **150**.

(31) In Step **604**, in a next recovery search, the wireless network module **150** detects power of all the frequencies or part of the frequencies of one or more bands to obtain a measurement result of frequencies/channels, wherein these frequencies/channels comprise at least part of standalone architecture (SA) FR1 and FR2 bands of the 5G system.

(32) In Step **606**, the wireless network module **150** determines the measurement result obtained in Step **604** is similar to the measurement result obtained in Step **602**, if yes, the flow enters Step **608**; and if not, the flow enters Step **610**. In this embodiment, the wireless network module **150** may use any suitable algorithm to determine the similarity between these two measurement results, such as calculating a distance between two measurement results, or using a suitable AI model to calculate a difference between two measurement results. Then, the wireless network module **150** can determine if the difference between two measurement results is within a range or lower than a threshold value to determine the similarity between these two measurement results. For example, if the difference between two measurement results is within the range or lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **604** is similar to the measurement result obtained in Step **602**; and if the difference between two measurement results is not within the range or not lower than the threshold value, the wireless network module **150** determines that the measurement result obtained in Step **604** is not similar to the measurement result obtained in Step **602**.

(33) In Step **608**, the wireless network module **150** determines that the UE **100** may be located in the same area without service, so the wireless network module **150** still disables the 5G function to avoid redundant cell search of the 5G system.

(34) In Step **610**, the wireless network module **150** determines that the UE **100** may be moved to an area with service, so the wireless network module **150** enables the 5G function, that is the wireless network module **150** can start to perform cell search of the 5G system.

(35) It is noted that the 5G system mentioned in FIG. **6** is for illustrative purposes only, not a limitation of the present invention. In other embodiments, the 5G function and the 5G system in FIG. **6** can be replaced by any suitable broadband cellular network technology system.

(36) Briefly summarized, in the control method of the UE of the present invention, by comparing the measurement results of the frequencies of one or more bands to determine a suitable system selection strategy, the UE can perform recovery search without doing some redundant steps, to effectively reduce the power consumption.

(37) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the

above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A control method of an electronic device, comprising: obtaining a first measurement result of frequencies of one or more bands, wherein the first measurement result comprises power of each frequency in at least part of the frequencies of the one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times, and the second measurement result comprises the power of each frequency in the at least part of the frequencies of the one or more bands; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result; wherein the first measurement result is generated when the electronic device is without wireless communication service and enters a first recovery search, and the second measurement result is generated when the electronic device is without wireless communication service and enters a second recovery search, wherein an execution time of the second recovery search is later than an execution time of the first recovery search; wherein the step of determining the system selection strategy according to the determination result comprises: in response to the determination result indicating that the second measurement result is similar to the first measurement result, using a first system selection strategy in the second recovery search or in a following recovery search; and in response to the determination result indicating that the second measurement result is not similar to the first measurement result, using a second system selection strategy in the second recovery search or in the following recovery search; wherein the step of using the first system selection strategy in the second recovery search or in the following recovery search comprises: not performing a cell search corresponding to the frequencies of the one or more bands in the second recovery search or in the following recovery search; and the step of using the second system selection strategy in the second recovery search or in the following recovery search comprises: performing the cell search corresponding to the frequencies of the one or more bands in the second recovery search or in the following recovery search.
2. The control method of claim 1, wherein the first measurement result and the second measurement result comprise power of each of the frequencies of the one or more bands.
3. A control method of an electronic device, comprising: obtaining a first measurement result of frequencies of one or more bands, wherein the first measurement result comprises power of each frequency in at least part of the frequencies of the one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times, and the second measurement result comprises the power of each frequency in the at least part of the frequencies of the one or more bands; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result; wherein the first measurement result is generated when the electronic device is without wireless communication service and enters a first recovery search, and the second measurement result is generated when the electronic device is without wireless communication service and enters a second recovery search, wherein an execution time of the second recovery search is later than an execution time of the first recovery search; wherein the step of determining the system selection strategy according to the determination result comprises: in response to the determination result indicating that the second measurement result is similar to the first measurement result, using a first system selection strategy in the second recovery search or in a following recovery search; and in response to the determination result indicating that the second measurement result is not similar to the first measurement result, using a second system

selection strategy in the second recovery search or in the following recovery search; wherein the step of using the first system selection strategy in the second recovery search or in the following recovery search comprises: not performing a public land mobile network (PLMN) and radio access technology (RAT) selection in the second recovery search or in the following recovery search; and the step of using the second system selection strategy in the second recovery search or in the following recovery search comprises: performing the PLMN and RAT selection in the second recovery search or in the following recovery search.

4. A circuitry of an electronic device, configured to perform the steps of: obtaining a first measurement result of frequencies of one or more bands, wherein the first measurement result comprises power of each frequency in at least part of the frequencies of the one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times, and the second measurement result comprises the power of each frequency in the at least part of the frequencies of the one or more bands; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result; wherein the first measurement result is generated when the electronic device is without wireless communication service and enters a first recovery search, and the second measurement result is generated when the electronic device is without wireless communication service and enters a second recovery search, wherein an execution time of the second recovery search is later than an execution time of the first recovery search; wherein the step of determining the system selection strategy according to the determination result comprises: in response to the determination result indicating that the second measurement result is similar to the first measurement result, using a first system selection strategy in the second recovery search or in a following recovery search; and in response to the determination result indicating that the second measurement result is not similar to the first measurement result, using a second system selection strategy in the second recovery search or in the following recovery search; wherein the step of using the first system selection strategy in the second recovery search or in the following recovery search comprises: not performing a cell search corresponding to the frequencies of the one or more bands in the second recovery search or in the following recovery search; and the step of using the second system selection strategy in the second recovery search or in the following recovery search comprises: performing the cell search corresponding to the frequencies of the one or more bands in the second recovery search or in the following recovery search.

5. The circuitry of claim 4, wherein the first measurement result and the second measurement result comprise power of each of the frequencies of the one or more bands.

6. A circuitry of an electronic device, configured to perform the steps of: obtaining a first measurement result of frequencies of one or more bands, wherein the first measurement result comprises power of each frequency in at least part of the frequencies of the one or more bands; obtaining a second measurement result of the frequencies of the one or more bands, wherein the first measurement result and the second measurement result are generated at different times, and the second measurement result comprises the power of each frequency in the at least part of the frequencies of the one or more bands; determining a similarity between the second measurement result and the first measurement result to generate a determination result; and determining a system selection strategy according to the determination result; wherein the first measurement result is generated when the electronic device is without wireless communication service and enters a first recovery search, and the second measurement result is generated when the electronic device is without wireless communication service and enters a second recovery search, wherein an execution time of the second recovery search is later than an execution time of the first recovery search; wherein the step of determining the system selection strategy according to the determination result comprises: in response to the determination result indicating that the second measurement result is similar to the first measurement result, using a first system selection strategy in the second recovery

search or in a following recovery search; and in response to the determination result indicating that the second measurement result is not similar to the first measurement result, using a second system selection strategy in the second recovery search or in the following recovery search; wherein the step of using the first system selection strategy in the second recovery search or in the following recovery search comprises: not performing a public land mobile network (PLMN) and radio access technology (RAT) selection in the second recovery search or in the following recovery search; and the step of using the second system selection strategy in the second recovery search or in the following recovery search comprises: performing the PLMN and RAT selection in the second recovery search or in the following recovery search.
